By JCX

Propty

The First AI-Backed Real Estate Company.

Smarter Search. Better Decisions. Bigger Future.

Product Deep Dive · Features, Tech Stack & Roadmap
In-depth specification for the engineering & product team

28 Features	10-Layer	6-Month	12 Sprints
Fully Specced	Tech Stack	Phased Roadmap	Acceptance Criteria

What's Inside

A	**Feature Architecture**	*Epic Map & Feature Inventory*
B	**Listing & Search Engine**	*Verified listings · filters · map search*
C	**Verification System**	*3-step badge · AI duplicate detection*
D	**AI Price Estimator**	*ML model · data pipeline · UX*
E	**Agent & Messaging**	*Profiles · chat · booking · ratings*
F	**Developer & Legal**	*Project portal · document tools*
G	**Commission Engine**	*Deal flow · payments · escrow*
H	**Admin & Analytics**	*Back-office · dashboards · moderation*
I	**Tech Stack Deep Dive**	*Layer-by-layer architecture*
J	**Data Model**	*Core entities & relationships*
K	**APIs & Integrations**	*3rd party services & endpoints*
L	**6-Month Roadmap**	*Sprint-level delivery plan*
M	**Roadmap Detail: Phase 0–1**	*Foundation & Core MVP*
N	**Roadmap Detail: Phase 2–3**	*Differentiation & Revenue*
O	**Risk Register**	*Impact · Mitigation · Owner*

A · FEATURE ARCHITECTURE

Epic Map & Feature Inventory

28 features across 8 epics — organised by user persona and delivery phase

🔍 **EP1**

Listing & Search

5 Features

Phase 1

5 features

✓ **EP2**

Verification

4 Features

Phase 1

4 features

🤖 **EP3**

AI Pricing

3 Features

Phase 2

3 features

💬 **EP4**

Agent & Messaging

5 Features

Phase 1

5 features

📅 **EP5**

Visit Booking

2 Features

Phase 2

2 features

🏗️ **EP6**

Developer Portal

3 Features

Phase 2

3 features

📄 **EP7**

Legal & Docs

3 Features

Phase 3

3 features

📊 **EP8**

Admin & Analytics

3 Features

Phase 2

3 features

P0 — Must have

P1 — Should have

P2 — Nice to have

Property Discovery: 5 Features

The core consumer loop — how users find, filter and shortlist properties

F-101 · P0 · Phase 1

P0

Multi-Parameter Search & Filter

Filters: city → area/thana → price range → bedrooms → bathrooms → sq ft range → property type (apartment/house/duplex/land/commercial) → furnishing → parking → floor → facing direction. Results update live (debounced 300ms). URL-serialisable filter state for sharing/bookmarking.

F-102 · P0 · Phase 1

P0

Map-Based Property Discovery

Mapbox GL JS integration. Property pins clustered at zoom-out. Click pin → mini-card with price, beds, verified badge. Draw polygon mode to search within custom area. Heatmap layer overlay for price density by area. Nearby amenities toggle (schools, hospitals, markets).

F-103 · P0 · Phase 1

P0

Listing Detail Page

Photo gallery (max 30 imgs, WebP auto-convert). Floor plan upload. Virtual tour embed (YouTube/Matterport). Full property spec table. Price per sq ft auto-calculated. Verified badge with audit trail. Share to WhatsApp/Facebook. Save to wishlist. View count (public). Contact CTA → in-app message or call-masked number.

F-104 · P1 · Phase 1

P1

Smart Search Suggestions & Bengali Support

Typeahead for area names in both English and Bengali script. Fuzzy matching (handles 'Gulshan' / 'gulsahn'). Recent searches saved. Popular searches surfaced. Address normalisation layer (Dhanmondi 27 vs Road 27 Dhanmondi handled as same). Bengali keyboard input supported.

F-105 · P1 · Phase 1

P1

Saved Searches & Alert Notifications

User saves a filter set as an 'Alert'. New matching listings trigger push notification (web) and email. Daily or instant frequency option. In-app notification centre. Alert management dashboard. Max 10 active alerts per free user, unlimited for Pro.

Trust Engine: 4 Features

How Propty eliminates fake listings — the core competitive moat

F-201 · P0 · Phase 1

P0

3-Step Listing Verification Workflow

Step 1 — Document Check: owner uploads title deed / allotment letter / rent deed. OCR extracts ownership name, matches against submitter account. Step 2 — Agent Confirmation: assigned agent confirms photos match physical property (GPS-tagged site photo required). Step 3 — Admin Sign-off: final QA. SLA: 48h standard, 24h express (paid). Badge types: Gold (all 3 steps), Silver (steps 1+2), Pending (in review).

F-202 · P0 · Phase 1

P0

AI Duplicate Detection

Before listing goes live, ML pipeline checks: (1) image perceptual hash comparison against all active listings in same area — flags visually duplicate photos. (2) NLP similarity on description text. (3) Address deduplication using PostGIS within 50m radius. Confidence score returned. >85% match → auto-flag for admin review. Prevents same flat being listed 10× by different agents.

F-203 · P1 · Phase 1

P1

Community Reporting & Badge Expiry

Any user can 'Report Listing' with reason (sold/rented, wrong photos, wrong price, doesn't exist). 3+ reports → auto-suspend pending admin review. Badge expiry: Gold badge expires 90 days after last confirmed availability. Owner must re-confirm listing is still available (one-click). Expired badges → downgrade to Silver or Pending.

F-204 · P1 · Phase 2

P1

Agent Identity Verification (KYC)

Agents must upload: NID (National ID), trade license or broker association membership, selfie photo. Verification via automated NID OCR cross-check + manual admin review. Verified agent badge on profile. Unverified agents limited to 5 listings. KYC renewal annually. Failed KYC → account suspended.

Propty Valuation Engine (PVE): 3 Features

Bangladesh's first ML-powered property price benchmark — kills opaque negotiation

PVE Model Architecture

Input Features

Area (sq ft), Location (lat/lng), Floor, Age, Furnishing, Type

Feature Encoding

One-hot: type/furnishing · Normalise: size/floor · PostGIS: proximity score

Model (Gradient Boost)

XGBoost trained on 50K+ BD transactions · Monthly retrain · MAE target <12%

Output

Price per sq ft · Confidence band (±10%) · Comparable listings shown

F-301 · P0 · Phase 2

P0

Buyer Price Check

On any listing page: 'Is this fairly priced?' → PVE returns: Fair / Overpriced / Good Deal badge + price range. Shows comparable sold/rented properties nearby with actual prices. Helps buyer negotiate.

F-302 · P1 · Phase 2

P1

Seller Price Guide

During listing submission: enter details → get suggested price range. Confidence score shown. Seller can override but flagged if >25% above PVE estimate (affects listing ranking).

F-303 · P2 · Phase 3

P2

Market Trend Dashboard

Public page showing price trends by area: avg price/sq ft last 12 months, YoY change, rental yield estimates. SEO-rich pages per area. Drives organic traffic.

Trust & Communication Layer: 5 Features

Replacing phone-number chaos with a structured, rated agent ecosystem

F-401 · P0 · Phase 1

P0

Agent Profile & Public Directory

Profile fields: photo, NID-verified badge, experience (years), speciality areas (Gulshan/Banani/Mirpur etc.), languages spoken, transaction count, avg response time, ratings breakdown (5-star). Public URL slug (proptybd.com/agent/[slug]). Sortable directory: by rating, area, response time. Free tier: 5 active listings. Pro tier: unlimited + analytics.

F-402 · P0 · Phase 1

P0

In-App Encrypted Messaging

WebSocket-based chat (Socket.IO). Conversation threads linked to specific listing. Pre-defined quick replies (schedule visit, send price, share docs). No phone numbers exposed until both parties agree. File sharing (PDF docs, photos). Message read receipts. Conversation archiving. Agent sees: lead source, listing viewed, user profile. User sees: agent response time badge.

F-403 · P0 · Phase 2

P0

Property Visit Scheduling

Calendar widget on listing page. Agent sets availability slots (Google Calendar / manual). Buyer picks slot. Confirmation SMS (BDApps / Twilio) + email. 24h reminder push notification. Post-visit: system prompts buyer to rate the visit and the agent. No-show tracking (3 no-shows = agent review). Visit history in user account.

F-404 · P1 · Phase 1

P1

Agent Rating & Review System

Post-deal or post-visit: 5-star rating + written review. Rating dimensions: accuracy of listing info, responsiveness, professionalism, area knowledge. Reviews moderated (profanity filter + admin flagging). Agent can respond to reviews (one reply per review). Aggregate score shown on profile & listing cards. Low-rated agents (< 3.0 avg) de-boosted in search.

F-405 · P1 · Phase 2

P1

Agent CRM Dashboard (Pro)

Pro agent features: Lead pipeline (New → Contacted → Visited → Offer → Closed). Lead source attribution (which listing, which channel). Follow-up reminders. Bulk messaging to saved leads (with opt-in). Monthly performance report (impressions, leads, conversion rate, revenue). Export to CSV.

B2B Layer + Legal Assistance: 6 Features

Bringing developers online + closing the legal gap that kills deals

F-501 · P1 · Phase 2

P1

Developer Company Page

Verified developer profile: company logo, founding year, total projects, REHAB membership badge. Project portfolio gallery. Buyer testimonials. SEO-optimised page per developer.

F-502 · P1 · Phase 2

P1

Project Listing (Under-Construction)

Multi-unit projects: floor plan per unit type, pricing table (from BDT X), progress photos with timestamp, expected handover date, payment plan (e.g. 20% down + installments), brochure PDF download, express interest CTA.

F-503 · P2 · Phase 3

P2

Developer Lead Portal

CRM for developer sales team: view interested buyers, contact info (after consent), source attribution, follow-up pipeline. Bulk update project status (construction update posts).

F-601 · P1 · Phase 2

P1

Bainanama Template Generator

Guided form: parties, property details, agreed price, advance paid, possession date → generates Bangla + English bainanama (sale agreement) PDF. Legally reviewed template. Not a substitute for lawyer but a solid starting draft.

F-602 · P1 · Phase 2

P1

Document Checklist & Tracker

Dynamic checklist per transaction type (purchase / rent / new project). Items: title deed, mutation (namjari), encumbrance certificate, NOC, tax clearance. Each item: status, upload slot, guidance note. Shareable checklist link for lawyer/agent.

F-603 · P2 · Phase 3

P2

Lawyer Referral Network

Curated network of verified property lawyers (Dhaka / CTG / Sylhet). Fixed fee packages: title search BDT 3K, deed registration support BDT 8K. Revenue share 15% to Propty. In-app booking + communication.

G+H · EP7+8 · COMMISSION ENGINE & ADMIN

Revenue Mechanics + Back-Office: 6 Features

How Propty captures value and keeps the platform healthy

F-701 · P0 · Phase 3

P0

Deal Declaration & Invoice Generation

When agent marks deal as 'Closed' in CRM: system auto-calculates commission (sale: 1-2% of price, rent: 0.5 month). Auto-generates invoice PDF. Sends to both parties. Commission amount locked — cannot be changed after declaration.

F-702 · P0 · Phase 3

P0

Payment Collection (bKash + SSLCommerz)

Primary: SSLCommerz (card/bank). Secondary: bKash merchant API. Tertiary: DBBL Nexus. Payment held in virtual escrow account (30-day hold). Dispute window: 15 days. If no dispute → auto-release to Propty. Retry logic for failed payments (3 attempts, 24h intervals).

F-703 · P1 · Phase 3

P1

Subscription Billing (Agent Pro)

BDT 500/month or BDT 4,500/year (25% saving). Auto-renew via bKash agreement / card token. Grace period: 7 days after failed payment before downgrade. Proration on mid-cycle upgrades. Invoice history in agent account.

F-801 · P0 · Phase 1

P0

Admin Moderation Panel

Internal tool (Next.js admin). Queues: verification review, KYC review, reported listings, disputed commissions. Bulk actions: approve/reject/suspend. Audit log on every action (who, when, what). Role-based access: Reviewer / Senior Admin / Super Admin.

F-802 · P1 · Phase 2

P1

Platform Analytics Dashboard

Internal metrics: DAU/MAU, listings by city, search queries heatmap, verification SLA compliance, agent response time, conversion funnel (search → view → contact → visit → deal). Exportable to CSV/PDF. Alerts: if DAU drops >20% day-on-day → Slack/email alert to team.

F-803 · P2 · Phase 3

P2

SEO & Content Automation

Auto-generate area landing pages: '2 BHK apartments for rent in Gulshan, Dhaka' — dynamic, updated daily. Schema.org markup for listings (RealEstateListing). Sitemap auto-generation. Canonical URL management. Bengali meta-tag support for BD Google Search.

Layer-by-Layer Architecture

10-layer stack chosen for BD developer talent availability, scalability & BD-specific integrations

01 Frontend

Next.js 14 + TypeScript + Tailwind CSS

SSR for SEO (property pages must rank). Strong BD dev talent. App Router for streaming.

02 Mobile (P2)

React Native (Expo) — Android-first

70%+ BD internet users on Android. Code reuse from Next.js logic layer. Expo OTA updates.

03 API Gateway

Node.js + Fastify + Zod validation

Fastify 2× faster than Express. Zod for runtime type-safety. BD talent pool large for Node.

04 Auth

Auth0 + Phone OTP (BDApps/Twilio)

Phone OTP critical — BD users register with mobile number. Auth0 for social login (Google/FB).

05 Database

PostgreSQL 16 + PostGIS extension

PostGIS enables geographic queries (listings within polygon, distance sort). ACID for financials.

06 Search

Meilisearch (self-hosted) → Elasticsearch

Meilisearch for Phase 1 speed/cost. Migrate to Elasticsearch at 100K+ listings for aggregations.

07 AI / ML

Python FastAPI + XGBoost + scikit-learn

PVE microservice. XGBoost outperforms deep learning on tabular property data (proven by Zillow).

08 Storage/CDN

AWS S3 + CloudFront CDN (Singapore region)

Singapore PoP: ~18ms latency to Dhaka. Auto-convert uploads to WebP (50% smaller). Signed URLs.

09 Payments

SSLCommerz + bKash Merchant API + DBBL Nexus

SSLCommerz is BD's dominant gateway. bKash: 70M+ users. DBBL for fallback. All BDT native.

10 Infra/DevOps

AWS ECS (Fargate) + RDS + GitHub Actions

Fargate: no server management. RDS for managed Postgres. GH Actions for CI/CD. Start small, scale.

Core Entities & Relationships

Simplified ERD — 8 core tables, their key fields and foreign key relationships

USERS
- id (uuid)
- name
- phone (unique)
- email
- role: buyer/agent/admin
- verified_at

LISTINGS
- id (uuid)
- owner_id → users
- agent_id → agents
- price, sq_ft
- lat/lng (PostGIS)
- status, badge

AGENTS
- id (uuid)
- user_id → users
- kyc_status
- avg_rating
- listing_count
- subscription_tier

MESSAGES
- id
- listing_id → listings
- sender_id → users
- receiver_id → users
- content, read_at
- attachment_url

VERIFICATIONS
- id
- listing_id → listings
- step1/2/3_status
- reviewed_by → users
- expires_at

DEALS
- id
- listing_id → listings
- buyer_id → users
- agent_id → agents
- agreed_price
- commission_amt, status

REVIEWS
- id
- deal_id → deals
- reviewer_id → users
- reviewee_id → users
- rating (1–5)
- body text

VISITS
- id
- listing_id → listings
- buyer_id → users
- agent_id → agents
- slot_start, slot_end
- status/no_show

Third-Party Services & Internal Endpoints

Every external dependency, what it does, and a fallback plan

Maps & Geo	**Mapbox GL JS**	Property pins, polygon search, heatmap *Fallback: Google Maps API (2× cost)*	Ph1
Search	**Meilisearch (self-hosted)**	Full-text listing search, Bengali support *Fallback: Elasticsearch SaaS*	Ph1
Auth	**Auth0**	Social login, JWT management *Fallback: Firebase Auth*	Ph1
SMS / OTP	**BDApps (Robi Axiata)**	Phone OTP, transactional SMS in BD *Fallback: Twilio (higher cost for BD)*	Ph1
Storage	**AWS S3 + CloudFront**	Property images, documents, floor plans *Fallback: Cloudflare R2 (cheaper egr*	Ph1
Email	**SendGrid**	Transactional email (verify, alerts, invoices) *Fallback: AWS SES (lower cost at sca*	Ph1
Payment (Primary)	**SSLCommerz**	Card, bank transfer, MFS payments *Fallback: SSLC has 99.5% uptime SLA*	Ph3
Payment (Mobile FS)	**bKash Merchant API**	bKash wallet payment, agreement-based billing *Fallback: Nagad / Rocket APIs*	Ph3
AI / OCR	**AWS Textract**	NID + title deed OCR for KYC/verification *Fallback: Google Vision API*	Ph2
Push Notifications	**Firebase FCM**	Web push + app push (visit reminders, alerts) *Fallback: OneSignal*	Ph1
CI/CD	**GitHub Actions + AWS ECS**	Build, test, deploy pipeline. Container orchestration *Fallback: Render.com for early MVP*	Ph1
Monitoring	**Sentry + AWS CloudWatch**	Error tracking, API latency, infra alerts *Fallback: Datadog (scale)*	Ph1

L · 6-MONTH ROADMAP

Sprint Delivery Overview

Month-by-month delivery across 4 phases — 2-week sprints, 12 sprints total

Phase 0 (Month 1) + Phase 1 (Month 2–3)

Foundation sprint + Core MVP sprint-by-sprint breakdown with acceptance criteria

PHASE 0 — FOUNDATION | Month 1 | Goal: Nothing ships to users. Everything ships to the team.

Sprint 1

- Design system: colour tokens, typography scale, component library (Figma + Storybook)
- Next.js 14 monorepo scaffold — app router, Tailwind, TypeScript strict mode
- PostgreSQL + PostGIS provisioned on AWS RDS. Schema v1 migrated (Liquibase)
- Auth0 tenant configured. Phone OTP flow via BDApps. Test login working
- GitHub Actions pipeline: lint → test → build → deploy to staging (ECS Fargate)

Sprint 2

- Fastify API: listing CRUD endpoints (POST /listings, GET /listings/:id, PATCH, DELETE)
- S3 bucket + CloudFront CDN configured. Image upload endpoint with WebP conversion
- Meilisearch instance deployed. Listing index synced from PostgreSQL (webhook on INSERT/UPDATE)
- Mapbox integration: basic map component with listing pins (lat/lng from DB)
- Admin panel scaffold (Next.js separate app). Auth0 role: admin. Listing review queue UI

PHASE 1 — CORE MVP | Month 2–3 | Goal: A working, live product with real listings in Dhaka.

Sprint 3–4 (Month 2)

- Listing submission flow: multi-step form (details → photos → docs → preview → submit)
- Verification queue in admin panel. Step 1 OCR (AWS Textract). Badge assignment logic
- Search page: filter sidebar + results grid + map view (toggle). URL-serialisable filters
- Listing detail page: photo gallery, spec table, verified badge, WhatsApp share, wishlist

Sprint 5–6 (Month 3)

- Agent profile pages. Agent directory (sortable). KYC upload flow. Verified agent badge

Phase 2 (Month 4–5) + Phase 3 (Month 6)

Differentiation & monetisation sprint breakdown with KPIs and go/no-go criteria

PHASE 2 — DIFFERENTIATION | Month 4–5 | Goal: 4 key differentiators live. 5,000 registered users.

Sprint 7–8 (Month 4)

- AI Price Estimator (PVE) microservice deployed (Python FastAPI + XGBoost). Beta label. Integrated on listing detail page
- Property visit booking: calendar widget, agent availability slots, SMS confirmation (BDApps), 24h reminder push
- Agent KYC: NID upload + AWS Textract + admin review queue. Verified agent badge logic
- Developer company page + project listing flow (multi-unit support, floor plan, payment plan)

Sprint 9–10 (Month 5)

- Featured listing monetisation: payment flow (SSLCommerz). Listing boost algorithm (featured score × recency)
- Bainanama template generator: form → PDF (pdfkit). Bengali + English output
- Document checklist tracker. Lawyer referral partner onboarding (target: 10 Dhaka lawyers by EOM)
- City rollout: seed 200+ listings in Chittagong, 100 in Sylhet. Partner with 20 agents per city
- Go/No-Go gate: ≥500 DAU, ≥2,000 verified listings, agent NPS > 30 → proceed to Phase 3

PHASE 3 — REVENUE & SCALE | Month 6 | Goal: First commission collected. 10,000 users. National coverage.

Sprint 11–12 (Month 6)

- Commission engine: deal declaration flow, invoice auto-generation, SSLCommerz + bKash payment, virtual escrow hold, 15-day dispute window, auto-release
- Agent Pro subscription: BDT 500/month billing, bKash agreement API, proration logic, grace period, downgrade on non-payment
- Analytics dashboard (internal): DAU/MAU, funnel conversion, verification SLA, agent performance. Sentry + CloudWatch alerts
- SEO automation: area landing pages auto-generated, sitemap, schema.org RealEstateListing markup
- National rollout: Rajshahi, Khulna, Sylhet seeded. Total target: 50 cities covered via partner agents
- KPIs at Month 6 end: 10,000 registered users · 10,000 live verified listings · 50 closed deals · BDT 500K+ commission collected

METRICS

How We Know Propty is Working

Phase-by-phase KPIs — what we measure, what targets we set, and what triggers a pivot

Phase 1
Month 2–3

Registered Users	
≥ 500 (beta)	≥ 1,000 (launch)
Live Verified Listings	
≥ 500 Dhaka	≥ 1,500 by EOM3
DAU / MAU	
–	≥ 200 DAU
Search → Listing Click CTR	
–	≥ 35%
Verification SLA (48h)	
–	≥ 90% on time
Agent Signups	
50 target	100 by EOM3

Phase 2
Month 4–5

Registered Users	
≥ 5,000	
DAU / MAU	
≥ 500 DAU	≥ 20% ratio
AI PVE Usage	
–	≥ 40% of detail page views
Visit Bookings	
≥ 50/month	
Developer Pages Live	
≥ 10	
Agent NPS	
–	≥ 30

Phase 3
Month 6

Registered Users	
≥ 10,000	
Live Verified Listings	
≥ 10,000	
Closed Deals	
≥ 50	
Commission Collected	
≥ BDT 500K	
Agent Pro Subscribers	
≥ 50	BDT 25K MRR
CAC (Cost per User)	
< BDT 500	

What Could Go Wrong & How We Respond

8 risks · Impact / Likelihood rating · Mitigation · Responsible owner

Cold Start — No listings at launch Mitigate: Pre-seed 500 verified listings in Dhaka with partner agents (incentive: free Pro 3 months). Launch invite-only with 200 beta users. Do not go public until ≥500 listings live.	Impact: High	Likelihood: High	*Owner: BD Ops Lead*
Agent commission resistance Mitigate: Never charge agents first. Free tier gives real value (leads, CRM, profile). Commission activates only on deal close. Agents earn more with Propty than without.	Impact: High	Likelihood: Med	*Owner: Product Lead*
Fake listings re-enter after verification Mitigate: Community reporting + 90-day badge expiry + spot re-verification (random 5% monthly). Image hash re-check on any photo update.	Impact: Med	Likelihood: High	*Owner: Trust & Safety*
AI price model inaccurate at launch Mitigate: Label as 'Propty Price Guide — not a valuation'. Grey-out PVE on areas with <20 data points. Improve model monthly as transactions accumulate.	Impact: Med	Likelihood: High	*Owner: ML Engineer*
bKash/SSLCommerz API downtime Mitigate: Multi-gateway fallback (SSLC → bKash → DBBL Nexus). Retry logic × 3. Webhook idempotency keys. Show user 'Try alternative payment method' UI instantly.	Impact: Med	Likelihood: Low	*Owner: Backend Lead*
Key competitor (bproperty) clones features Mitigate: Data moat (verified listing DB + transaction history) is hard to clone. Speed matters more than secrecy. Ship and iterate.	Impact: Med	Likelihood: Med	*Owner: CEO / PM*
BD regulatory change on property platforms Mitigate: Consult REHAB (Real Estate & Housing Association of BD) early. Register as marketplace, not broker. Legal retainer from day one.	Impact: High	Likelihood: Low	*Owner: Legal Counsel*
Dev team scope creep / timeline slippage Mitigate: PM gates every sprint with acceptance criteria (defined in this doc). Phase gates with measurable go/no-go criteria. No feature added to a sprint without PM + Tech Lead sign-off.	Impact: High	Likelihood: Med	*Owner: PM + Tech Lead*

Propty

By JCX · Beyond Boundaries

The First AI-Backed Real Estate Company.

AI understands data. We understand you.

28	10	12	8
Features Fully Specced	Tech Stack Layers	Sprints 6-Month Plan	Epics End-to-End

Next Steps

1. PM + Tech Lead review this doc. Lock feature scope for Sprint 1.
2. Recruit: 2× Next.js, 1× Node.js, 1× Python ML, 1× UI/UX (all Dhaka-based).
3. Procure: Auth0, BDApps OTP, Mapbox, SendGrid, AWS. Budget: ~$300/month Phase 1.
4. Partner outreach: 50 Dhaka agents before soft launch. Incentive: 3 months free Pro.